EXCELLENCE

CLASS 11 - 2026-27

Physics | Motion 02

Motion in a Straight Line

Distance & Displacement Deep Dive · Arcs · x–t Graphs

By - Excellence Programme

20 October 2026

What we learn today

01 Lock the may / must logic for distance & displacement

02 More 1-D, 2-D and 3-D path calculations

03 Square, hexagon and triangle geometry shortcuts

04 Motion on a circular arc — distance and chord

05 Reading x–t graphs (graph is not the path)

06 Board MCQs from today's ideas

Link from Motion 01

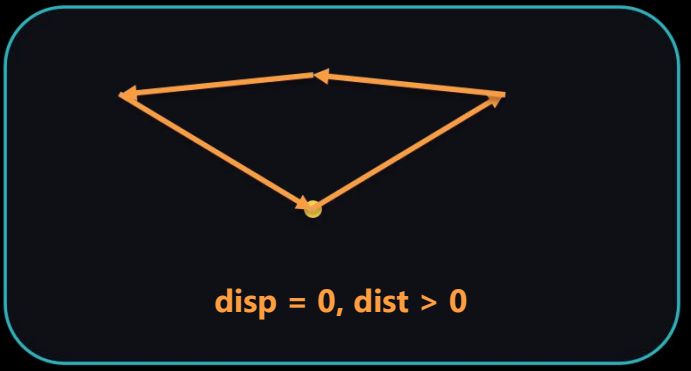
Last class we met frame, rest–motion, and the basic meaning of distance and displacement. Today we go deeper: harder paths, polygon shortcuts, circular arcs, and graphs.

$\Delta r = r_f - r_i$ | distance = length of the actual path

Same unit (metre) does not mean same quantity. Distance is scalar. Displacement is vector.

May / must logic — lock this table

Statement	Truth
Distance = 0 ⇒ displacement = 0	MUST (object did not move)
Displacement = 0 ⇒ distance = 0	MAY / MAY NOT (round trip)
Displacement ≠ 0 ⇒ distance = 0	MUST NOT
Distance ≠ 0 ⇒ displacement = 0	MAY / MAY NOT
In every 1-D motion, distance = |disp|	FALSE (turns break equality)



MCQ — two statements

Stmt-1: Distance and displacement are different quantities.

Stmt-2: They have the same unit and same dimension.

1. Only Stmt-1 is correct
2. Both are correct
3. Only Stmt-2 is correct
4. Both are incorrect

Solution: MCQ — two statements

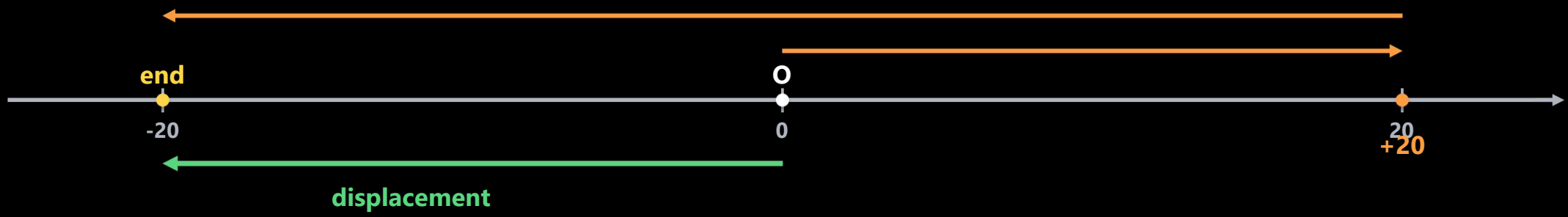
Idea: different nature (scalar vs vector), but both measured in metre with [L].

Answer: both are correct

Rough work / extra work

1-D: out and past the origin

On the x-axis: start at origin, go to +20 m, then reverse and stop at −20 m.

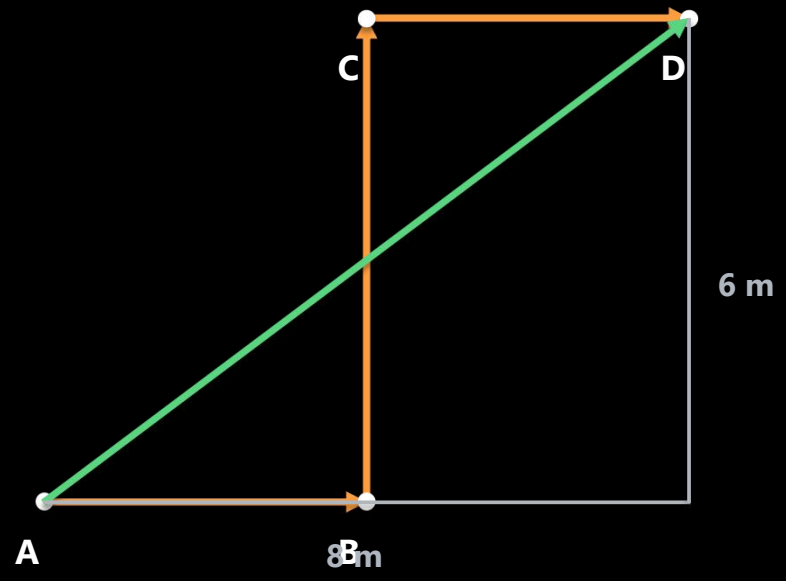


distance = 20 + 40 = 60 m

displacement = (−20)î − 0 = −20 î m

L-path: 4 m + 6 m + 4 m

A → B = 4 m right, B → C = 6 m up, C → D = 4 m right.



distance = 14 m

$|\Delta r| = \sqrt{(8^2 + 6^2)} = 10$ m

14 m , 10 m

MCQ — L-path numbers

For the path above, distance and |displacement| are

1. 14 m, 8 m
2. 14 m, 6 m
3. 14 m, 10 m
4. None

Solution: MCQ — L-path numbers

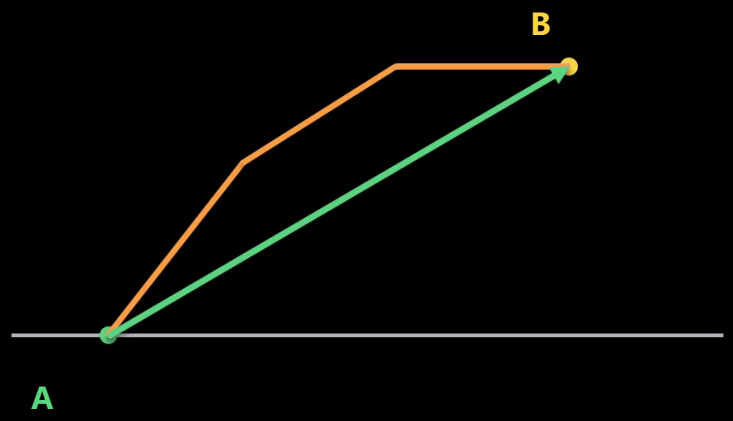
Step 1: distance = 14 m , |Δr| = 10 m

Answer: option 3

Rough work / extra work

Endpoints only — distance may be unknown

Object goes from (0, 0) to (2, −4) on a curved path. Without the path, distance cannot be found. Displacement still can.

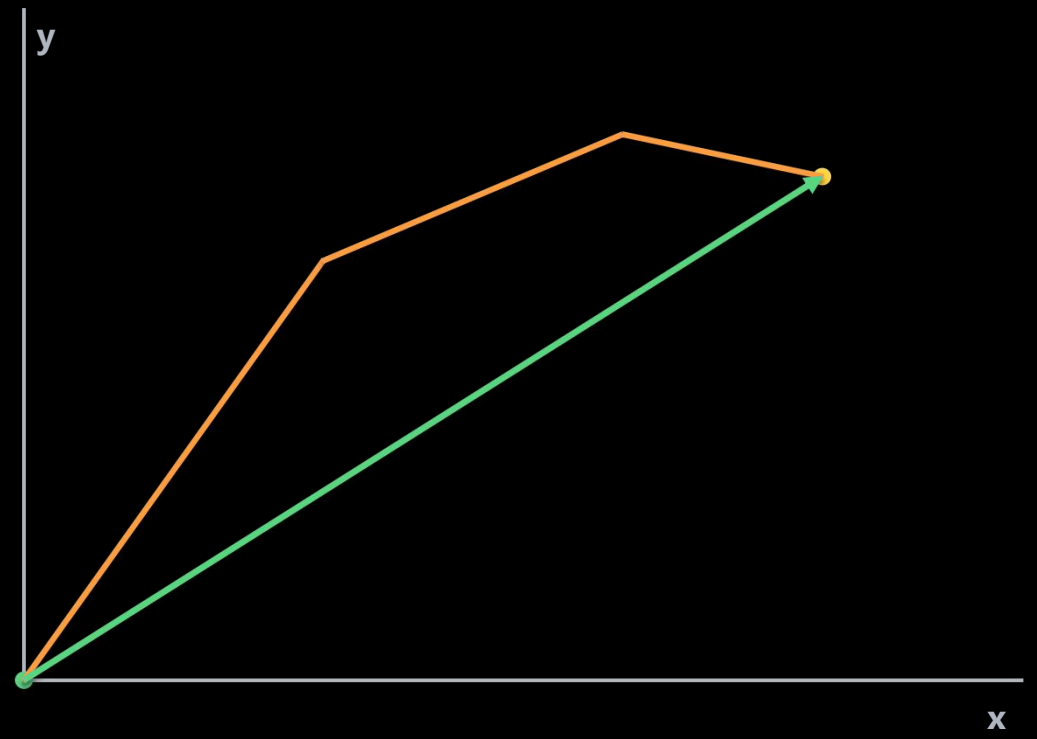


$\Delta \vec{r} = 2\hat{i} - 4\hat{j}$, $|\Delta \vec{r}| = 2\sqrt{5}$ m

Endpoints alone → displacement yes, distance maybe no.

Projectile landing & 3–4–5

Ends at (8, 6) after a curved flight from the origin. Path length unknown; net shift is a 3–4–5 style triangle ×2.



$\Delta \mathbf{r} = 8\hat{\imath} + 6\hat{\jmath}$

$|\Delta \mathbf{r}| = \sqrt{(64+36)} = 10\ \text{m}$

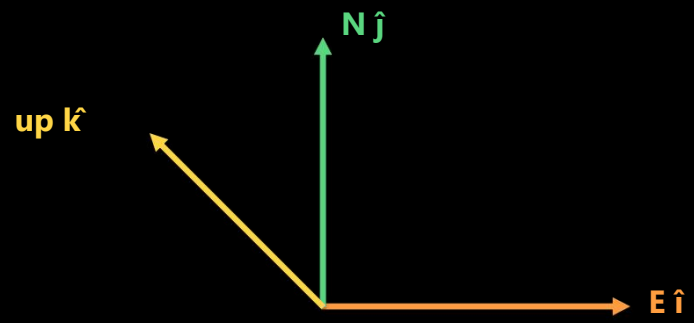
3-D walk: North, East, Up

Walk 10 m North, 10 m East, 10 m vertically up.

distance = 10 + 10 + 10 = 30 m

$\Delta r = 10\hat{i} + 10\hat{j} + 10\hat{k}$

$|\Delta r| = 10\sqrt{3}$ m



MCQ — P(2, 3, 5) → Q(3, 4, 5)

Displacement vector is

1. $\hat{i} + \hat{j} + 10\hat{k}$
2. $\hat{i} + \hat{j} + 5\hat{k}$
3. $\hat{i} + \hat{j}$
4. $2\hat{i} + 4\hat{j} + 6\hat{k}$

Solution: MCQ — P(2, 3, 5) → Q(3, 4, 5)

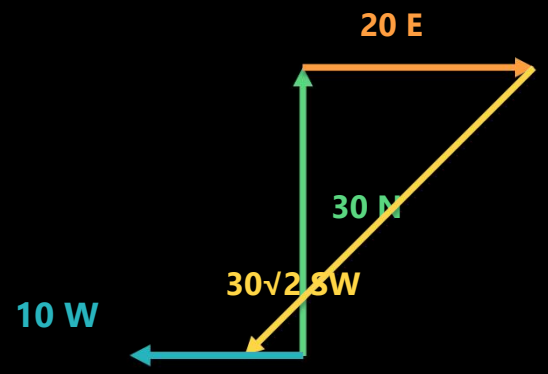
Step 1: $\Delta \vec{r} = (3-2)\hat{i} + (4-3)\hat{j} + (5-5)\hat{k} = \hat{i} + \hat{j}$

Answer: $\hat{i} + \hat{j}$

Rough work / extra work

IIT-style walk: N, E, then SW

30 m North, 20 m East, then 30√2 m South-West. Find displacement.



$\text{SW} = 30\sqrt{2} \times (-\hat{i} - \hat{j})/\sqrt{2} = -30\hat{i} - 30\hat{j}$

$\text{Net} = 20\hat{i} + 30\hat{j} - 30\hat{i} - 30\hat{j} = -10\hat{i}$

Displacement = 10 m West

MCQ — IIT walk answer

Net displacement in the walk above is

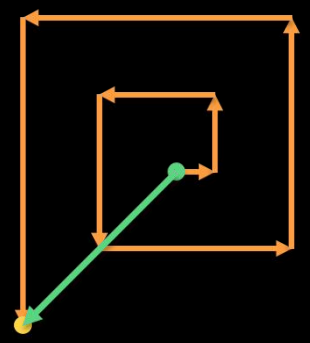
1. 10 m East
2. 10 m North
3. 10 m West
4. 10√2 m South-West

Solution: MCQ — IIT walk answer

Answer: 10 m West

Spiral segments 1…8 m

Alternate right, up, left, down with lengths 1, 2, 3, 4, 5, 6, 7, 8 m.



distance = 36 m

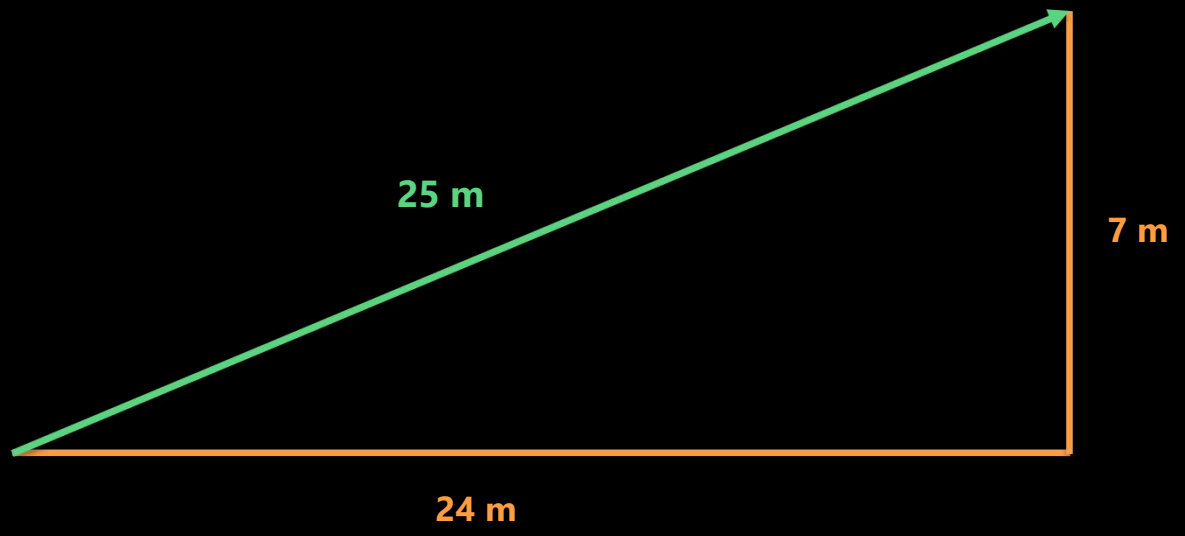
$\Delta r = -4\hat{i} - 4\hat{j}$

$|\Delta r| = 4\sqrt{2}$ m

Staircase path → 7–24–25

Horizontal pieces add to 24 m. Vertical pieces add to 7 m. Net is a right triangle.

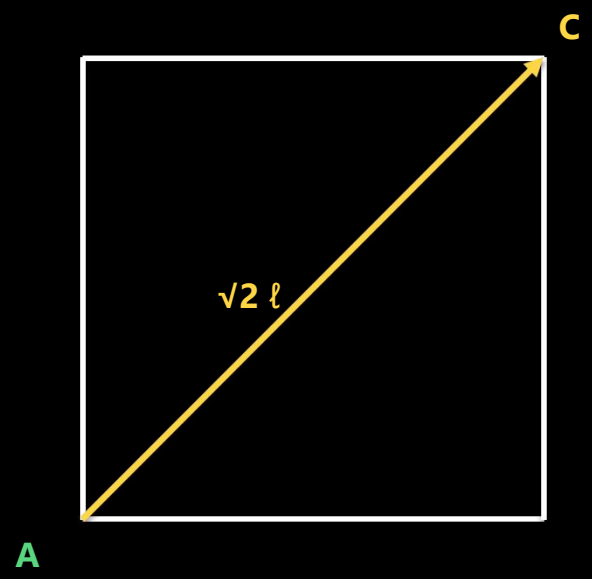
$$|\Delta \mathbf{r}| = \sqrt{(24^2 + 7^2)} = \sqrt{(576+49)} = \sqrt{625} = 25 \text{ m}$$



Rough work / extra work

Square park shortcuts

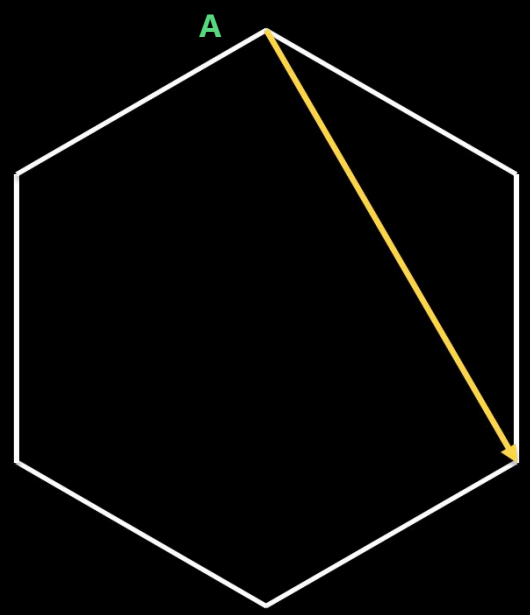
Square ABCD of side ℓ. Start at A. Walk on the boundary.



Path	distance	|disp|
A → B	ℓ	ℓ
A → C via B	2ℓ	$\ell\sqrt{2}$
A → D via B, C	3ℓ	ℓ
A → A lap	4ℓ	0

Regular hexagon — distance vs chord

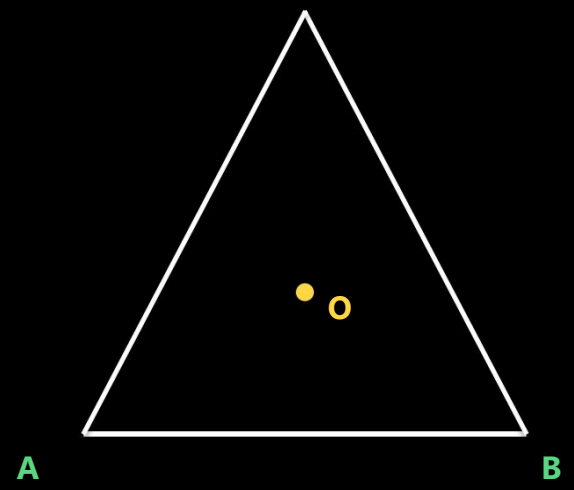
Regular hexagon side ℓ. From vertex A, walking around the boundary.



Sides walked	distance	|disp|
1	ℓ	ℓ
2	2ℓ	$\ell\sqrt{3}$
3	3ℓ	2ℓ
6 (full)	6ℓ	0

Equilateral triangle + centroid

Equilateral triangle side ℓ. Centroid O. Useful lengths: vertex→centroid = $\ell/\sqrt{3}$; centroid→side midpoint = $\ell/(2\sqrt{3})$.

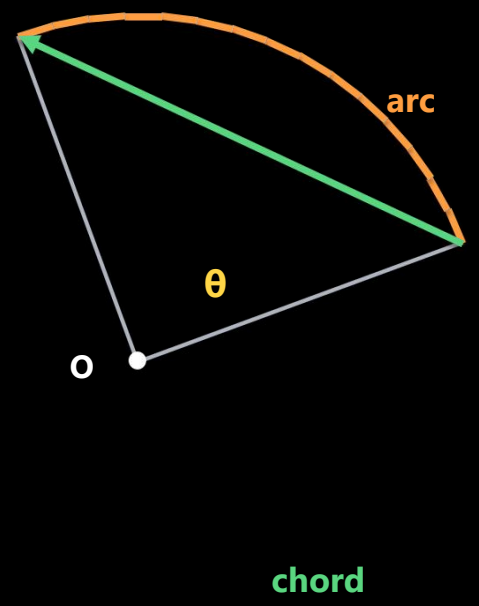


Example A → B → O: distance = $\ell + \ell/\sqrt{3}$, |disp| = $\ell/\sqrt{3}$

Example O → midpoint → B: use $\ell/(2\sqrt{3}) + \ell/2$ for distance

Circular arc — the two lengths

Object moves on a circle of radius R through angle θ (in radian). Distance follows the arc. Displacement is the chord.



distance = R θ | |displacement| = 2 R sin(θ/2)

Why the chord formula works

Join O to the ends A and B. Triangle OAB is isosceles with OA = OB = R and angle AOB = θ. Drop a perpendicular from O to chord AB — it bisects θ and the chord.

Half chord = R sin(θ/2)

Full chord = |Δr| = 2 R sin(θ/2)

Put θ in radian for distance = Rθ. For 60° use θ = π/3, not 60.

Rough work / extra work

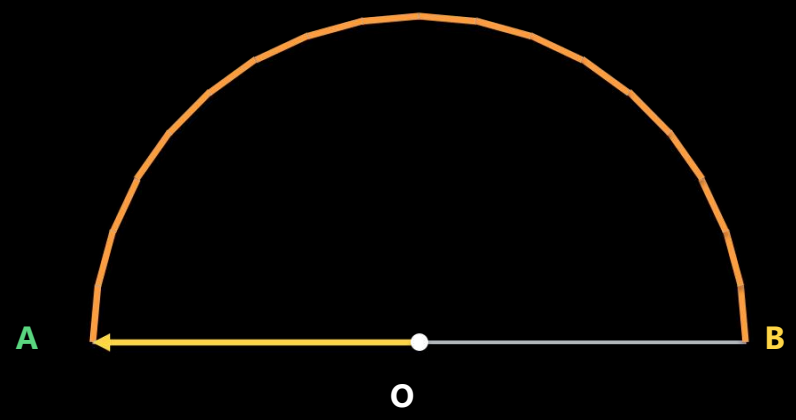
Arc angles you must own

θ	distance Rθ	|disp| = 2R sin(θ/2)
60° = π/3	R π/3	R
90° = π/2	R π/2	R√2
120° = 2π/3	2πR/3	R√3
180° = π	π R	2R
270° = 3π/2	3πR/2	R√2

Small θ: sin(θ/2) ≈ θ/2 ⇒ |disp| ≈ Rθ = distance (almost equal).

Radial then semicircle

From centre O go out along a radius to A, then along the upper semicircle to B (opposite end of diameter).



distance = R + π R

displacement = R (from O to B along diameter)

dist = R(1+π) , |disp| = R

3.5 revolutions on a circle

Period for one lap = 40 s. Radius R. After 140 s the object has completed 140/40 = 3.5 revolutions.

distance = 3.5 × 2πR = 7π R

After half-integer turns, ends at the opposite point on diameter

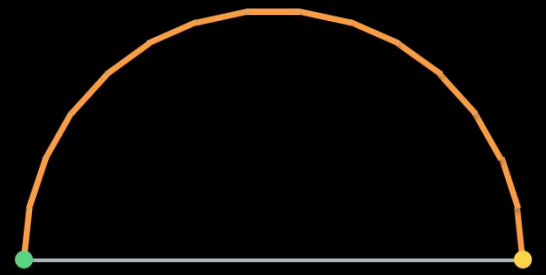
|displacement| = 2R

ratio distance / |disp| = 7π / 2

MCQ — circle $(x-5)^2 + y^2 = 25$

Particle goes along the upper semicircle from (0, 0) to (10, 0). Distance is

1. 10
2. 5π / 2
3. 5π
4. 10π



Solution: MCQ — circle $(x-5)^2 + y^2 = 25$

Idea: centre (5, 0), R = 5. Upper semicircle arc = $\pi R = 5\pi$.

Answer: 5π

Rough work / extra work

x–t graph is not the path

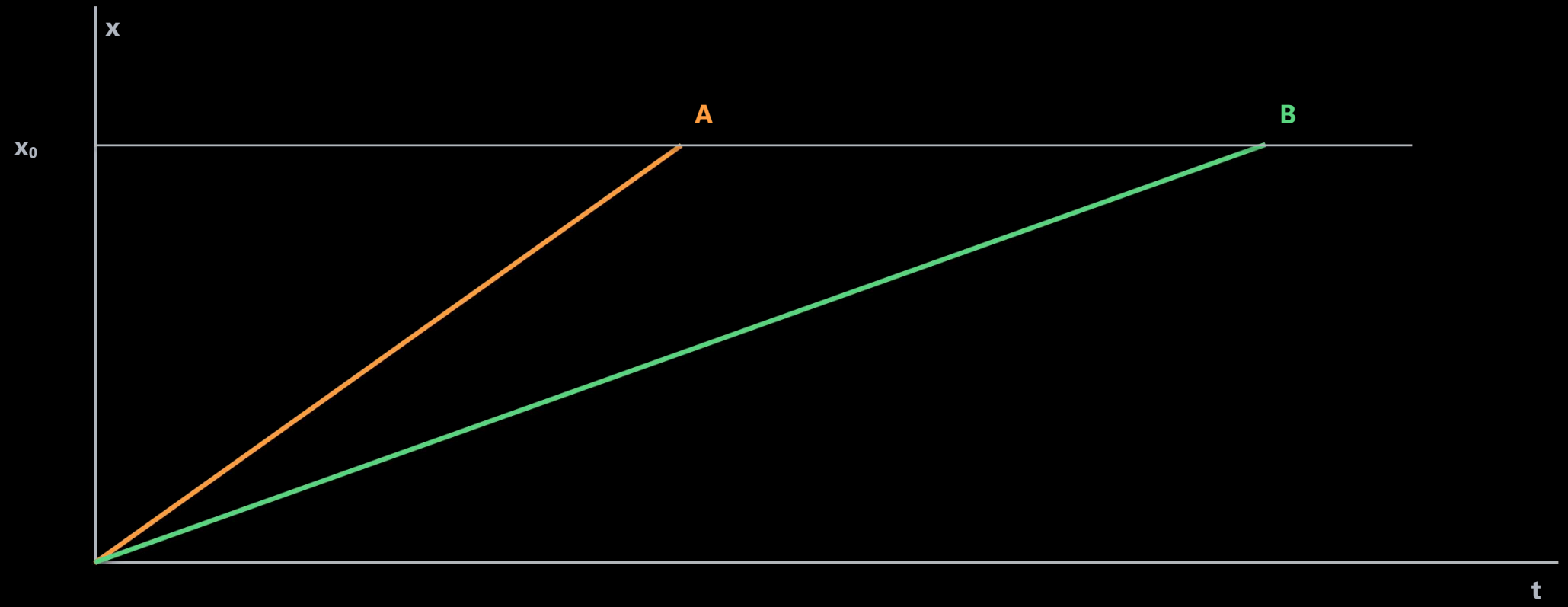
A position–time graph plots x against t. The curve on paper is not the road the object walked. Read change in x for displacement. Add |Δx| of each piece for distance when it turns.

displacement = $x_f - x_i$ | distance = sum of |Δx| on each leg

A sloping line on x–t means uniform velocity — not a sloping road.

Two travellers, same Δx

A reaches x_0 at time t_1. B reaches the same x_0 at later time t_2. Both start from x = 0.



MCQ — same final x

For A and B above, which is correct?

1. displacement of A = displacement of B
2. distance of A = distance of B always
3. A and B have equal average speeds
4. A's distance is always greater

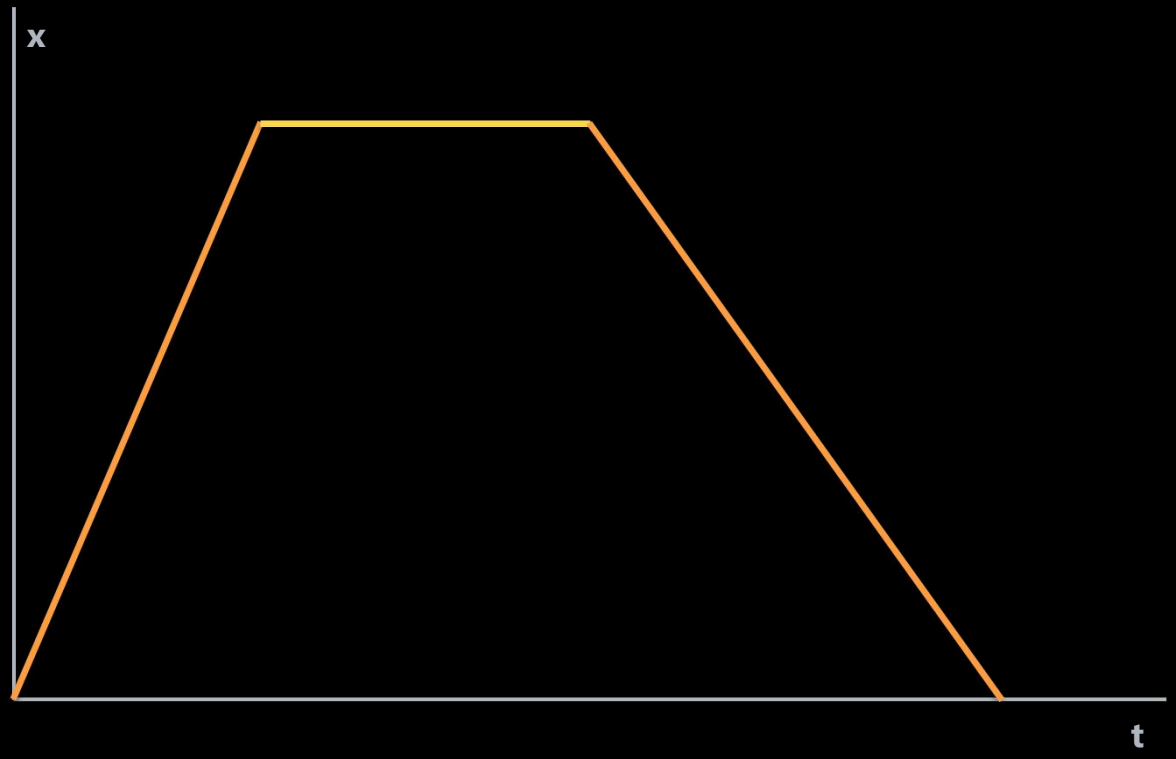
Solution: MCQ — same final x

Idea: both go 0 → x_0 with no turn, so Δx is the same. Speeds differ because × differ.

Answer: displacement of A = displacement of B

x–t with rest and return

Read carefully: go forward, rest (horizontal), then return.



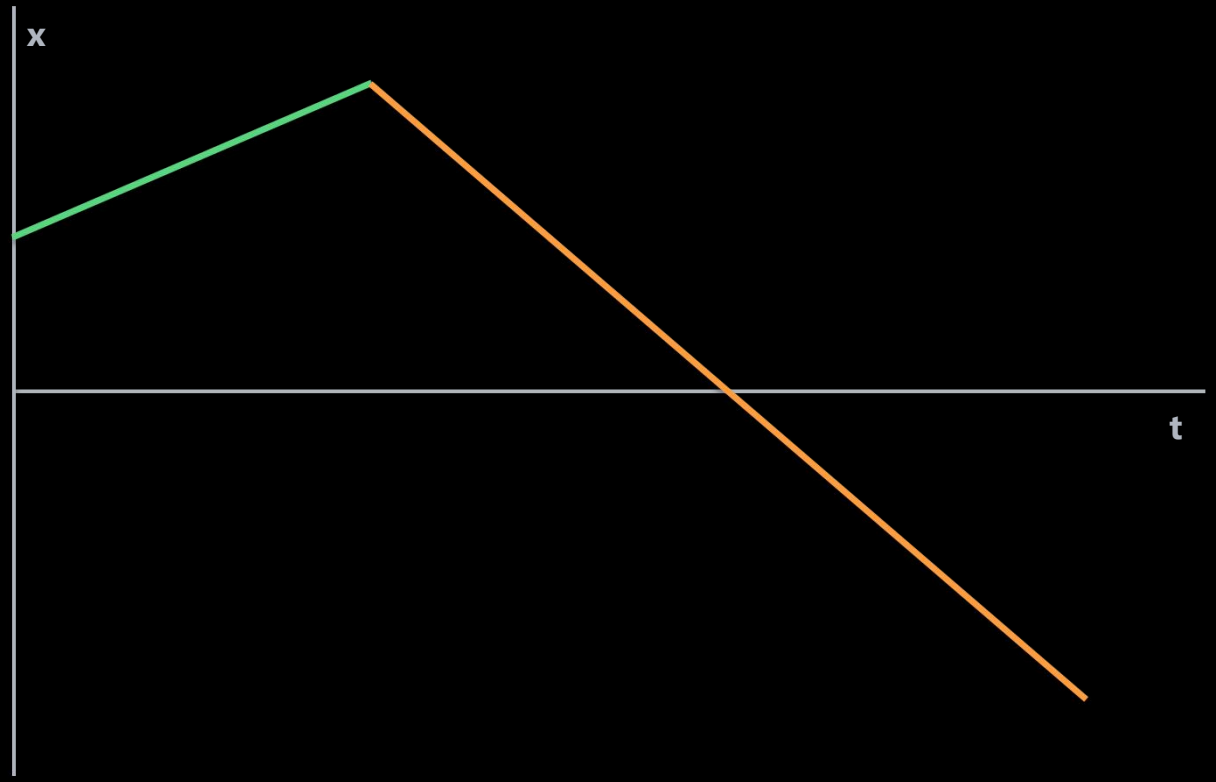
displacement = 0 − 0 = 0

distance = 10 + 0 + 10 = 20 m

Back to start → disp 0, distance 20 m

Rough work / extra work

x–t: 10 → 20 → −20



displacement = (−20) − 10 = −30 m

distance = 10 + 40 = 50 m

Distance adds the absolute climbs.
Displacement only cares about start and end x.

Valid distance–time graph

Distance never decreases with time for a real moving clock. A distance–time graph cannot go down, cannot be negative, and cannot show two distances at one time.

Reject
Decreasing d–t (distance fell)

Reject
Negative distance

Reject
Two values at one time

Accept
Non-decreasing $d \geq 0$

MCQ — which d–t is valid?

Which distance–time graph can be correct?

1. A graph that decreases after some time
2. A graph that goes below d = 0
3. A vertical line (two d at one t)
4. A non-decreasing graph with d ≥ 0

Solution: MCQ — which d–t is valid?

Answer: non-decreasing graph with d ≥ 0

Displacement–time can go down

Displacement (or position) can increase or decrease. An s–t or x–t graph may slope up or down. That is allowed — the object can reverse.

- Distance–time: never decreases
- Displacement–time: may decrease (sign / direction change)

Do not mix the two graph rules.

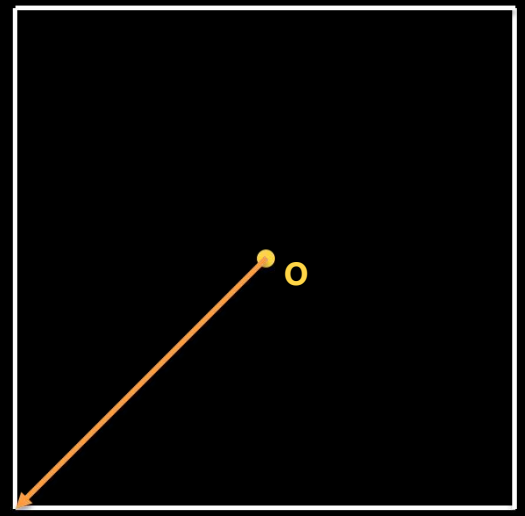
Rough work / extra work

Square from centre O

From centre O of a square side ℓ, go to a corner, then walk three sides to the opposite corner B.

distance = $\ell/\sqrt{2} + 3\ell$

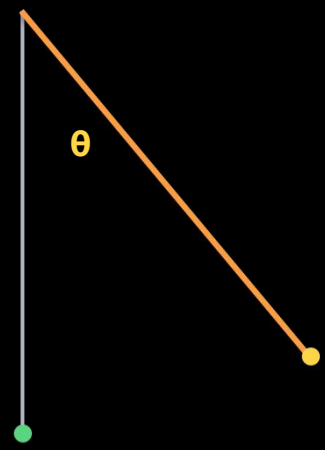
|displacement| = $\ell/\sqrt{2}$ (O to opposite corner is also half-diagonal)



Pendulum release to lowest point

Bob released from angle θ with vertical length ℓ. Ends at the bottom (ℓ, 0) if we set coordinates from the support carefully — use start (ℓ cosθ, ℓ sinθ) → (ℓ, 0) in the usual school sketch.

$$\Delta \mathbf{r} = \ell(1 - \cos\theta)\,\hat{\imath} - \ell \sin\theta\,\hat{\jmath}$$



Say these aloud

- Distance = path sum. Displacement = $r_f - r_i$.
- Same unit ≠ same quantity (scalar vs vector).
- Circle: distance = $R\theta$, $|\text{disp}| = 2R \sin(\theta/2)$.
- Hexagon / square chords beat long perimeter walks.
- x–t is not the road — read Δx carefully.
- d–t never falls; s–t / x–t may fall.

Next: speed and velocity (PW 03 / 04).

Thank you

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